Robot Design Workshop!

Today's Plan

Intro

10 min

Design activity

35 min

Interview + Survey

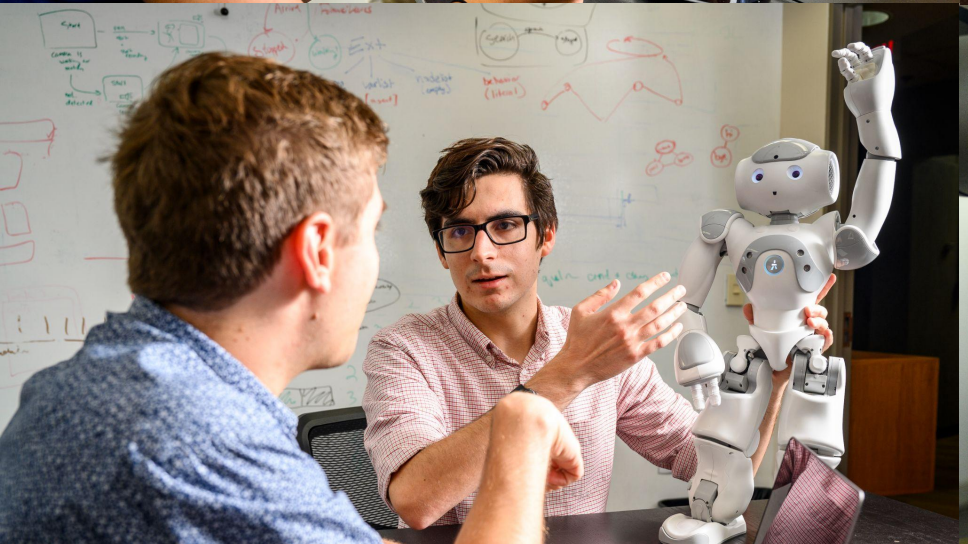
15 min

What will we do today?

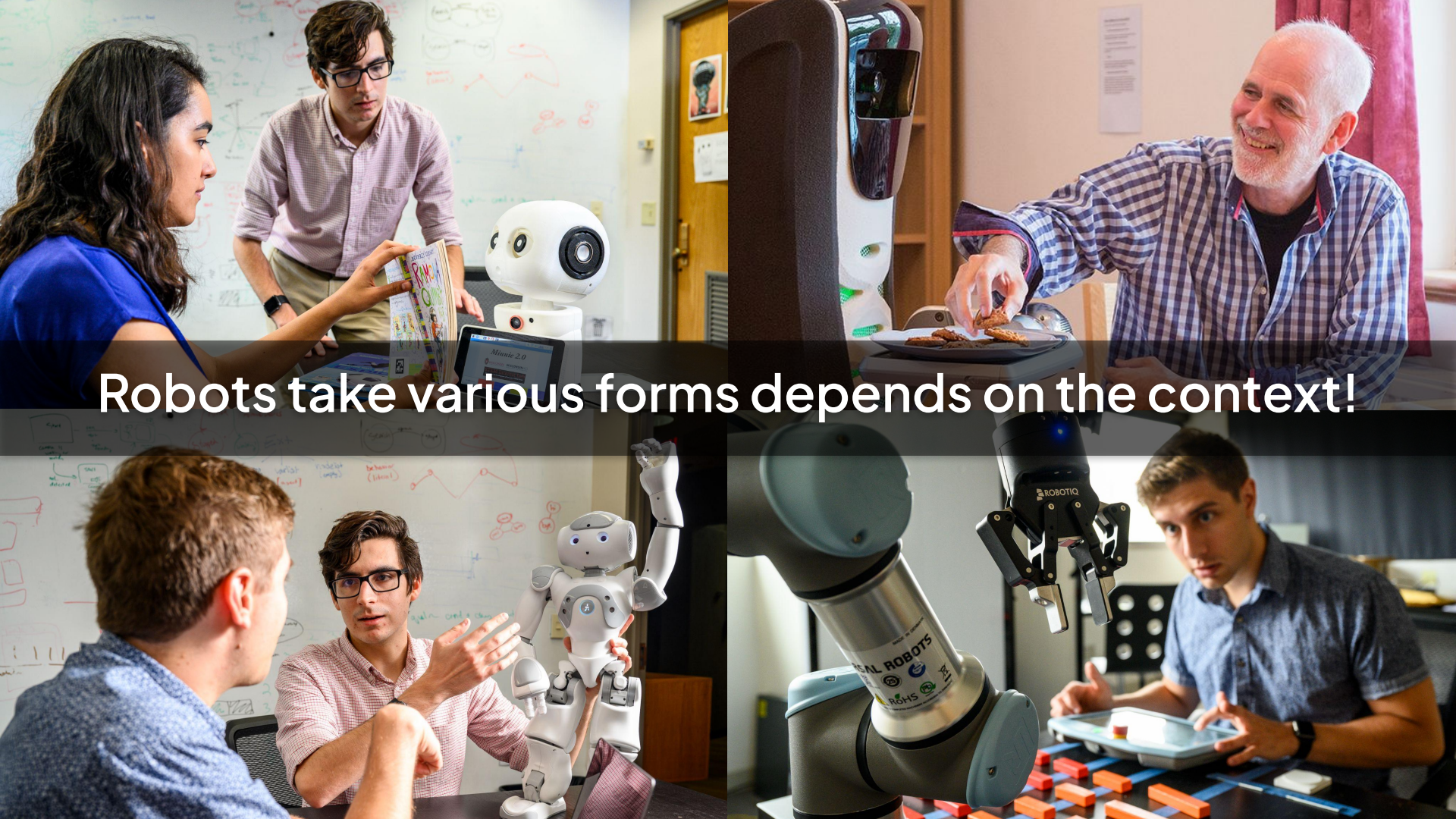
We will *design* a robot!

Oh? *Robot*? 🤖

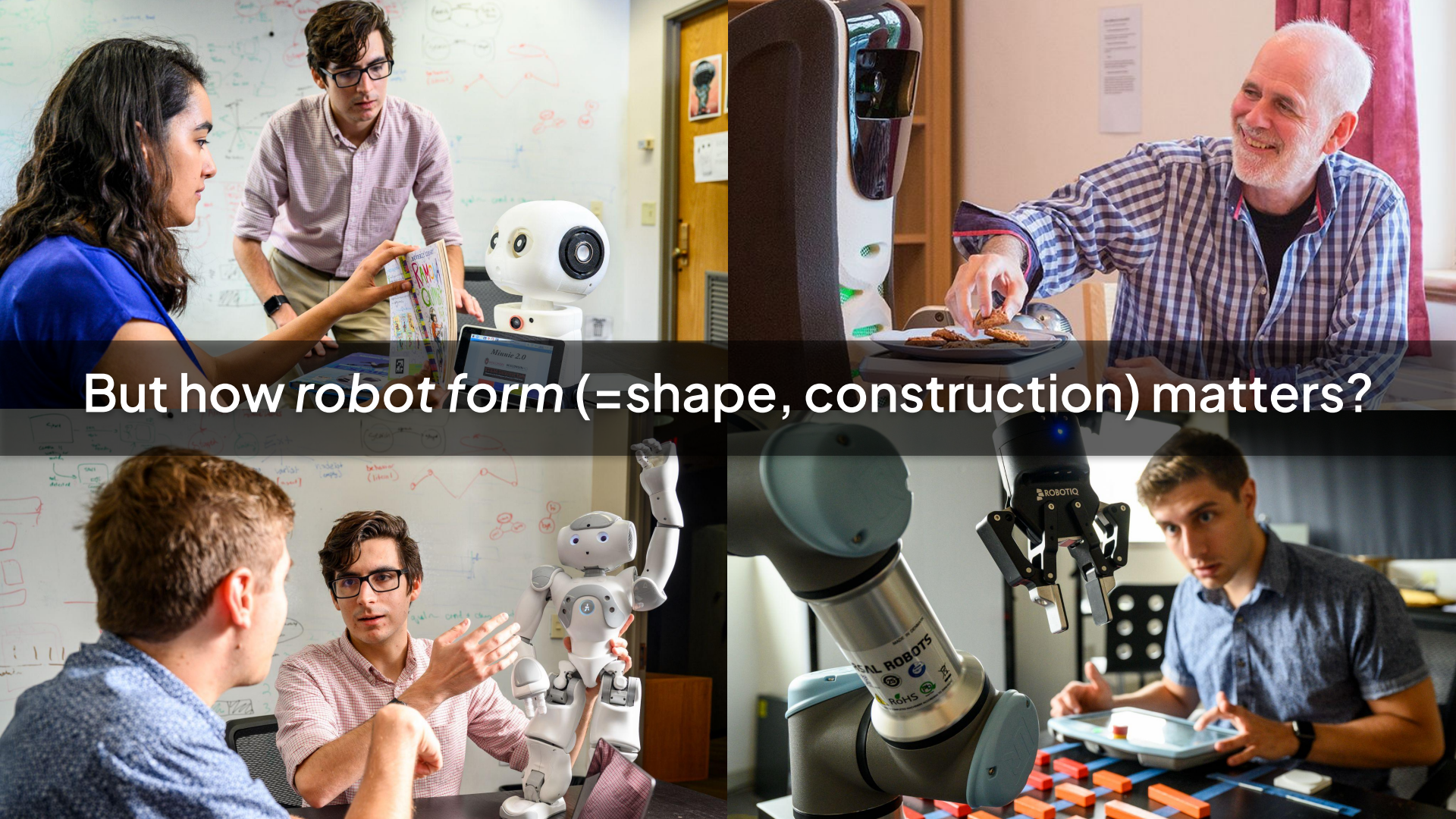
Minnie 2.0



ROBOTIQ
ROHS



Robots take various forms depends on the context!



But how *robot form* (=shape, construction) matters?

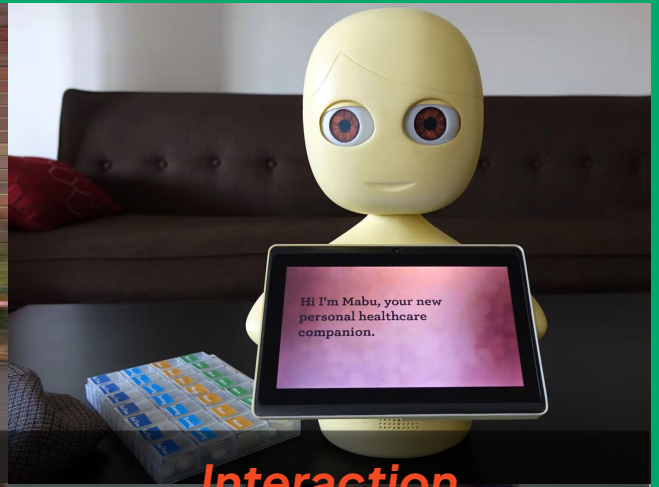
enable locomotion...

Function

enable movement

enable locomotion...

give focus of attention...



Function

Interaction

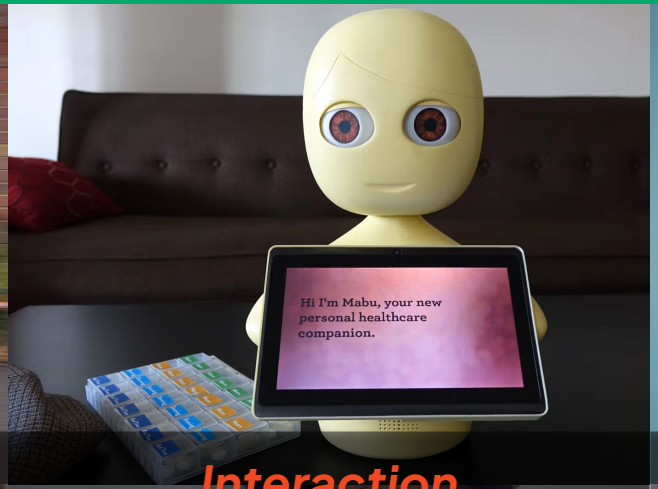
enable movement

imply proximity...

enable locomotion...

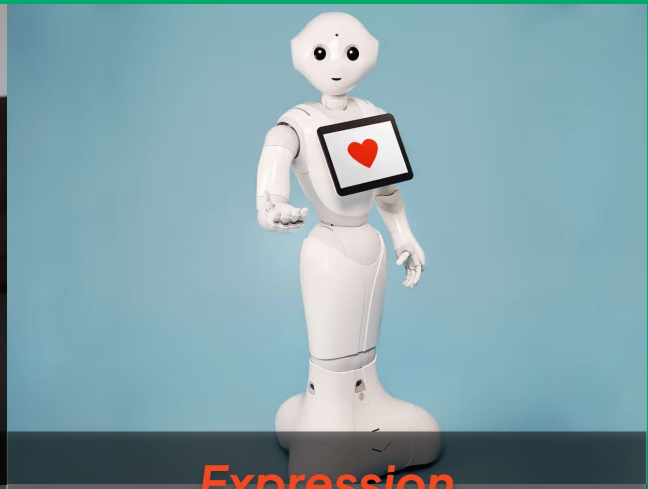
Function

give focus of attention...



Interaction

create impression...



Expression

enable movement

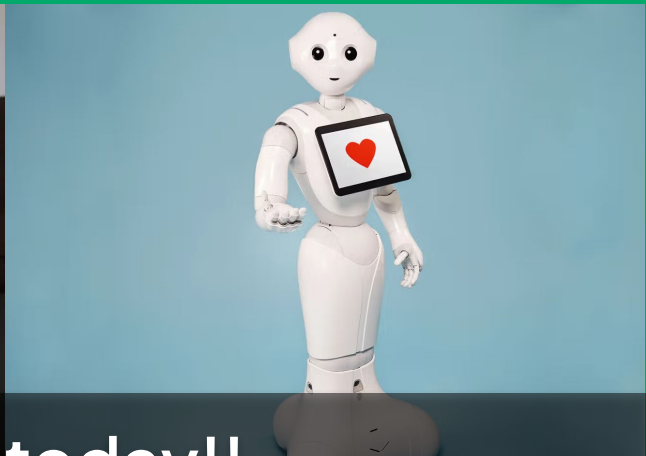
imply proximity...

convey its presence...

enable locomotion...

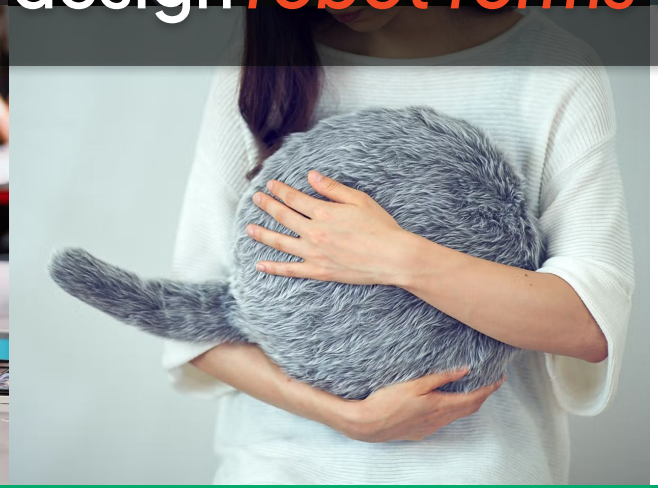
give locus of attention...

create impression...

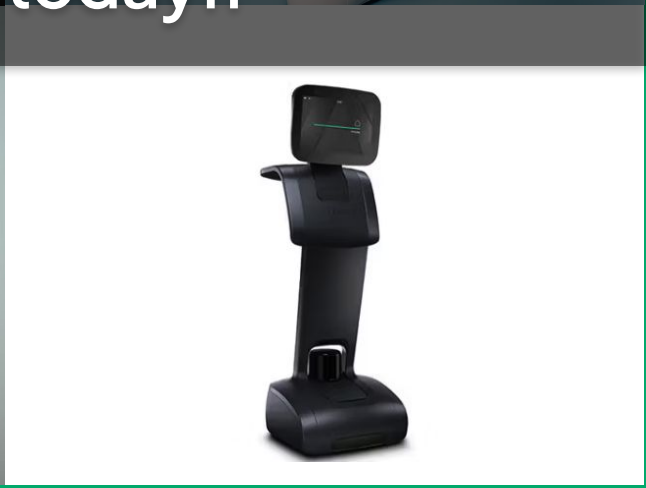


We will design *robot forms* today!!

enable movement



imply proximity...



convey its presence...

Your Turn!

Will repeat as time permits!

Step 1

Concept Making

Brainstorm, sketching, etc.

Step 2

Form Exploration

Use blocks!!

Step 3

Finalize

Finalize your design!

Step 1: Concept Making!

3 mins

1. Pick one context from the list below that interests you
2. Use the worksheet to jot down ideas or make sketches, think about what the robot might be like…

A. **Design a patrol robot that can navigate a crowded public space.**

B. **Design a delivery robot that can handle fragile objects in the outdoor environment.**

C. **Design an education assistant robot that can support learning in a classroom environment.**

Step 2-1: Pick Keywords

1 mins

1. Refer back to your concept from the previous step
2. Pick one keyword card from the list that best connect to your concept (or that just caught your eyes!)
3. Get one keyword card from the facilitator
4. Make sure you have two keywords at this point!!

Machinelike	Mechanical	Unfriendly	Unintelligent	**Anxious*** * you feel anxious towards robot
Humanlike	Organic	Friendly	Intelligent	**Relaxed*** * you feel relaxed towards robot

Step 2-2: Form Exploration!

15 mins

1. Familiarize yourself with the blocks!
2. Design one robot for each keyword using the blocks. (You will build 2 robots in total)
3. Use blocks as many(few) as you want. + Feel free to use clay if you want to create any elements/components.
4. Let the facilitator know when you feel done! Will take photos.

Step 3: Finalizing!

5 mins

1. Finalize the robots you designed in the previous step by answering the following questions. Use worksheets!
2. Questions:
   a. *What is the name of the robot?*
   b. *What would the size, color, texture or material of the robot be ?*
   c. *How would the robot behave? (e.g., how would the robot interact with people or environment?)*
3. Let the facilitator know when you feel done! We'll do quick reflection.